Lab Summary: Decentralized Trust & P2P Protocols

1 Key Learning Objectives

By the end of this lab sequence, students will be able to:

1. **Evaluate Reputation Vulnerabilities:** Identify structural weaknesses in endorsement-based trust systems, specifically the mechanics and impact of **Sybil Attacks**.
2. **Assess Authentication Controls:** Distinguish between identity creation (registration) and identity verification (authentication), understanding why digital signatures are essential for immutable logs.
3. **Analyze Routing Efficiency:** Compare the performance trade-offs between Unstructured P2P networks (Gossip/Flooding) and Structured Overlays (DHTs), specifically analyzing bandwidth usage vs. lookup speed.
4. **Navigate Distributed Hash Tables:** Understand the mechanics of DHTs and be able to manually conduct and debug DHT routing alogorithms.

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2 Lab Module 1: Distributed Trust & Reputation

This module focuses on the application layer, exploring how peers establish trust in a decentralized environment without a central authority to vet users.

2.1 Activity A: The Sybil Attack (Identity Inflation)

- **Scenario:** Students interact with a deployed "Endorsement-Based Reputation System." The system calculates trust scores based on how many other organizations endorse a user.

- **The Incident:** Students troubleshoot reports of low-quality actors having artificially high reputation scores.
- **The Exploit:** Students discover that the system allows unrestricted registration of new "Organizations." They demonstrate a **Sybil Attack** by scripting the creation of hundreds of fake entities to cross-endorse a specific target, inflating its score to the top of the leaderboard.
- **Insight:** Students learn that in open systems, "one identity, one vote" is impossible to enforce without proof-of-humanity or cost-of-entry mechanisms.
- **Objective Alignment:** Supports *Objective 1* by demonstrating how cheap identity creation undermines reputation metrics.

2.2 Activity B: Broken Authentication (Spoofing)

- **Scenario:** Students are given a "Patched" version of the previous system. The "Open Registration" loophole has been closed (registration is now restricted or costly).
- **The Incident:** Despite the fix, reputable organizations claim they are being listed as endorsing projects they have never heard of.
- **The Exploit:** Students analyze the endorsement packet structure and realize that while *registration* is secure, the *endorsement messages* lack cryptographic signatures. Students demonstrate an **Impersonation Attack** by injecting an endorsement packet claiming to be from a trusted node without possessing that node's private key.
- **Insight:** Students learn the difference between **Authorization** (Who is allowed to be here?) and **Authentication** (Did they actually say this?).
- **Objective Alignment:** Supports *Objective 2* by highlighting the necessity of Public Key Infrastructure (PKI) and digital signatures.

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3 Lab Module 2: P2P Overlays & Routing

This module moves to the network layer, contrasting "Unstructured" networks (Gossip) with "Structured" networks (DHTs).

3.1 Activity A: Manual Chord Lookup (The Finger Table)

- **Exercise:** Students are provided with a visual representation of a **Chord Ring** and a specific node's **Finger Table**.
- **The Task:** Students must manually trace the look-up path for a specific file key (ID). They must identify which "Finger" allows them to jump halfway around the ring, and iteratively narrow down the distance to the target node.

- **Insight:** Students visualize the **O(log N)** efficiency of DHTs, seeing how a lookup takes only a few hops even in a large keyspace, compared to a linear search.
- **Objective Alignment:** Supports *Objective 4* by solidifying the concept of logarithmic routing geometry.

3.2 Activity B: Unstructured P2P (Gossip vs. Flooding)

- **Simulation:** Students interact with a simulated mesh network lacking a structured index (Gossip Protocol). They attempt to locate a specific file using two methods:
    1. **Random Walk:** Students query neighbors sequentially. They observe that this method is extremely slow and often fails to find the file before timeout (ineffective).
    2. **Flood Search:** Students send a broadcast message to all peers, who forward it to their peers.
- **Outcome:** The file is found almost instantly, but students observe a massive spike in network traffic metrics.
- **Insight:** Students recognize the trade-off: **Flooding** is fast but creates a bandwidth storm (not scalable), whereas **Random Walks** are low-bandwidth but unreliable. This motivates the need for Structured Overlays (DHTs).
- **Objective Alignment:** Supports *Objective 3* by contrasting the "Blind Search" of unstructured networks with the "Directed Search" of DHTs.

3.3 Activity C: Advanced Debugging (The Pastry Freeze)

- **Simulation:** Students run a stress-test on a buggy Python implementation of a **Pastry DHT** (Structured Overlay).
- **The Bug:** The implementation relies exclusively on **Prefix Routing**. When routing to a key that does not correspond to an exact Node ID, the logic hits a "hole" in the routing table and fails.
- **The Fix:** Students analyze the node state to realize that while the *Routing Table* had no path, the **Leaf Set** contained a candidate closer to the destination.
- **Objective Alignment:** Supports *Objective 4* by forcing students to reconcile the dual-nature of DHTs: utilizing shortcuts for speed but relying on ring neighbors for guaranteed delivery.